Shor's Algorithm on $2n + 3$ Qubits

Architectural Blueprint and State-Vector Analysis of a Generalized Qiskit Implementation

Beauregard (2003) architecture with Draper QFT arithmetic

Companion document to `shor_core_generalized.ipynb`

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Scope and conventions

This document maps the exact quantum state, in Dirac notation, at every stage of the generalized circuit implemented in `shor_core_generalized.ipynb`, and proves why each stage does what it claims. The circuit factors any odd composite N that is not a prime power, with any base a satisfying $1 < a < N$ and $\gcd(a, N) = 1$. No problem-specific shortcut is used: modular multiplication is built from Draper's Fourier-space adders [3], assembled into Beauregard's $2n + 3$ qubit circuit [2].

Notation.

- $n = \lceil \log_2(N + 1) \rceil$ is the bit length of N, so $N < 2^n$; $m = n + 1$.
- $r = \mathrm{ord}_N(a)$ is the smallest $r > 0$ with $a^r \equiv 1 \pmod N$. The circuit never uses r.
- t is the counting precision and $M = 2^t$. The default is $t = 2n$.
- $\mathrm{e}(\theta) := e^{2\pi i\theta}$. Note $\mathrm{e}(\theta + k) = \mathrm{e}(\theta)$ for every integer k.
- Registers are little-endian, as in Qiskit: a k-qubit register in state $|v\rangle$ holds $v = \sum_{i=0}^{k-1} v_i 2^i$, and qubit 0 is the least significant bit.
- Gates: H, X, $P(\theta) = \mathrm{diag}(1, e^{i\theta})$, controlled phase $CP(\theta)$, CX, controlled swap $CSWAP$. $CP(\theta)$ multiplies $|11\rangle$ by $e^{i\theta}$ and is symmetric in its two qubits.

Table 1 links every symbol below to the notebook function that builds it.

Symbol	Notebook function	Action
QFT_m, $\mathrm{QFT}_m^\dagger$	`qft_noswap`, `iqft_noswap`	Fourier representation $\phi(b)$, section 3.1
$\Phi\mathrm{ADD}(a)$	`phi_add`	$\phi(b) \mapsto \phi(b + a)$
C-$\Phi\mathrm{ADD}(a)$	`c_phi_add`	controlled add
CC-$\Phi\mathrm{ADD}(a)$	`cc_phi_add`	doubly controlled add
CC-$\Phi\mathrm{ADD}(a)\mathrm{MOD}(N)$	`cc_phi_add_mod`	controlled modular add, section 3.4
C-$\mathrm{MULT}(a)\mathrm{MOD}(N)$	`c_mult_mod`	$\lvert x\rangle \lvert b\rangle \mapsto \lvert x\rangle \lvert (b + ax) \bmod N\rangle$
C-U_a	`controlled_ua`	$\lvert x\rangle \lvert 0\rangle \mapsto \lvert ax \bmod N\rangle \lvert 0\rangle$
$\mathrm{QFT}_t^\dagger$ (counting)	`iqft_counting`	interference, section 4
whole algorithm	`build_shor_circuit`	modes `semiclassical` and `full`
$P(y)$ closed form	`exact_distribution`	ideal output distribution, section 4.3
continued fractions	`continued_fraction`, `convergents`	section 5.3

Table 1: Symbols and their implementation.

1 Resource analysis and initialization

1.1 Qubit allocation

The notebook offers two modes that produce the same output distribution (theorem 4.4). The `semiclassical` mode is Beauregard's $2n + 3$ qubit architecture; the `full` mode keeps an explicit t-qubit counting register and is the form in which the linear algebra is easiest to read.

Why b has $n + 1$ qubits. Inside the modular adder the register holds $b + a \in [0, 2N - 2]$ and $b + a - N \in [-N, N - 2]$. Both fit in $n + 1$ bits: the first because $2N - 2 < 2^{n+1}$, the second in two's complement because $N < 2^n$. The top qubit b_n is therefore an overflow and sign bit: after the inverse Fourier transform, $b_n = 1$ exactly when the stored value is negative.

Why one ancilla. The modular adder must remember whether $b + a < N$ in order to add N back; `anc` stores that bit and is returned to $|0\rangle$ before the adder ends (theorem 3.5).

Register	Role	`semiclassical`	`full`	Initial state
`ctrl` / `count`	phase-estimation control	1	t	$\lvert 0\rangle$ / $\lvert 0\rangle^{\otimes t}$
`x`	work, holds $a^x \bmod N$	n	n	$\lvert 1\rangle_n$
`b`	Fourier accumulator	$n+1$	$n+1$	$\lvert 0\rangle_{n+1}$
`anc`	modular-adder ancilla	1	1	$\lvert 0\rangle$
Total		$\mathbf{2n+3}$	$t+2n+2 = 4n+2$	

Table 2: Qubit allocation as a function of the bit length n of N (with $t = 2n$ in `full` mode).

Why $t = 2n$. The continued-fraction step is guaranteed to recover s/r whenever $M \geq N^2$ (theorem 5.1); $t = 2n$ gives $M = 2^{2n} > N^2$.

1.2 Gate counts

Write $m = n+1$. Counting the gates emitted by the notebook functions (exact QFTs; "$\leq$" because phase gates whose angle is a multiple of 2π are not emitted):

Block	CP	CX	P	H	other
QFT_m or $\mathrm{QFT}_m^\dagger$	$m(m-1)/2$			m	
$\Phi\mathrm{ADD}(a)$			$\leq m$		
C-$\Phi\mathrm{ADD}(a)$	$\leq m$				
CC-$\Phi\mathrm{ADD}(a)$	$\leq 3m$	2			
CC-$\Phi\mathrm{ADD}(a)\mathrm{MOD}(N)$	$\leq 2m^2+8m$	8	m	$4m$	$2\,X$
C-$\mathrm{MULT}(a)\mathrm{MOD}(N)$	$\leq n(2m^2+8m)+m(m-1)$	$8n$	nm	$4nm+2m$	$2n\,X$
C-U_a	$\leq 2n(n+1)(2n+11)$	$16n$	$2nm$	$8nm+4m$	n CSWAP, $4n\,X$

Table 3: Logical gate counts of the building blocks.

The full algorithm applies up to $t = 2n$ controlled multipliers, so the exact-QFT circuit uses $O(n^4)$ two-qubit gates. With the approximate QFT [5] of degree k (notebook setting `AQFT_MAX_K`), each Fourier transform costs $O(mk)$ instead of $O(m^2)$, the modular adder costs $O(nk)$, a controlled multiplier $O(n^2k)$, and the whole circuit $O(n^3k)$. With $k = O(\log n)$ this is the $O(n^3 \log n)$ bound of Beauregard [2]. The semiclassical counting side adds $t(t-1)/2$ classically controlled phases, $2t$ Hadamards, t measurements and $t-1$ resets.

Table 4 lists the exact logical counts produced by the notebook (with `SKIP_IDENTITY = True`: a controlled multiplier whose constant $a^{2^j} \bmod N$ equals 1 is the identity and is omitted).

N	a	mode	qubits	t	nontrivial C-U	CP	CX	CSWAP
15	7	semiclassical	11	8	2	1394	128	8
15	7	full	18	8	2	1422	128	8
21	2	semiclassical	13	10	10	11160	800	50
21	2	full	22	10	10	11205	800	50
35	3	semiclassical	15	12	12	21699	1152	72

Table 4: Exact logical gate counts before transpilation.

Remark 1.1. For comparison, a special-purpose $N = 15$ circuit that exploits $15 = 2^4 - 1$ (multiplication by 2^k as a bit rotation) needs only 36 two-qubit gates and 5 CSWAPs. The general construction costs about forty times more for the same N. This is the price of correctness for arbitrary N, and it is the dominant factor in the hardware benchmarks.

Remark 1.2 (Approximate QFT inside the adders). Dropping rotations $\pi/2^j$ with $j > k$ reduces the CP count for $N = 21$, $a = 2$ from 11160 to 9840 ($k = 3$), 8520 ($k = 2$) and 6760 ($k = 1$). The arithmetic then becomes approximate. Measured against the exact distribution, the Hellinger

fidelity of the ideal output is 1.000, 0.99998, 0.99966 for $N = 15,\ t = 8$ and 0.990, 0.953, 0.935 for $N = 21,\ t = 5$ (for $k = 3, 2, 1$).

1.3 Initial state

$$|\psi_0\rangle^{\text{full}} = |0\rangle_{\text{count}}^{\otimes t} \otimes |1\rangle_x \otimes |0\rangle_b \otimes |0\rangle_{\text{anc}}\,, \qquad |\psi_0\rangle^{\text{semi}} = |0\rangle_{\text{ctrl}} \otimes |1\rangle_x \otimes |0\rangle_b \otimes |0\rangle_{\text{anc}}\,. \tag{1}$$

The work register starts at $|1\rangle$ (one X gate on x_0) because every later operation multiplies it: $1 \cdot a^x = a^x$. Starting from $|0\rangle$ would give 0 forever.

2 Superposition

In `full` mode a Hadamard acts on each counting qubit. Since $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$,

$$H^{\otimes t}|0\rangle^{\otimes t} = \bigotimes_{j=0}^{t-1} \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{M}} \sum_{x_{t-1},\ldots,x_0 \in \{0,1\}} |x_{t-1}\cdots x_0\rangle = \frac{1}{\sqrt{M}} \sum_{x=0}^{M-1} |x\rangle\,, \tag{2}$$

because expanding the product picks one term from each factor, and every t-bit string appears exactly once. Therefore

$$|\psi_1\rangle = \frac{1}{\sqrt{M}} \sum_{x=0}^{M-1} |x\rangle\,|1\rangle_x\,|0\rangle_b\,|0\rangle_{\text{anc}}\,. \tag{3}$$

The amplitude $1/\sqrt{M}$ is fixed by normalisation, $M \cdot (1/\sqrt{M})^2 = 1$.

Example 2.1. For $t = 2$: $\frac{1}{2}(|0\rangle+|1\rangle)(|0\rangle+|1\rangle) = \frac{1}{2}(|00\rangle+|01\rangle+|10\rangle+|11\rangle) = \frac{1}{2}(|0\rangle+|1\rangle+|2\rangle+|3\rangle)$.

In `semiclassical` mode only one qubit is put into $|+\rangle = (|0\rangle+|1\rangle)/\sqrt{2}$ at a time, once per step. The superposition over all x is never stored at once, yet the measurement statistics are identical (section 4.4). The full-mode state above is the reference picture for the analysis.

3 Entanglement: generalized modular exponentiation

The goal of this stage is the map

$$|x\rangle\,|1\rangle_x\,|0\rangle_b\,|0\rangle \;\longmapsto\; |x\rangle\,|a^x \bmod N\rangle_x\,|0\rangle_b\,|0\rangle \tag{4}$$

for all x simultaneously, built bottom-up from additions that are performed as phase rotations.

3.1 The Fourier representation

Lemma 3.1 (Fourier representation without swaps)**.** *Let* QFT_m *be the circuit that, for* $i = m-1, m-2, \ldots, 0$*, applies* H *to qubit* i *and then* $CP(\pi/2^{i-l})$ *between qubit* l *and qubit* i *for every* $l < i$*. Then for every* $b \in [0, 2^m)$

$$\phi(b) := \mathrm{QFT}_m\,|b\rangle = \bigotimes_{i=0}^{m-1} \frac{|0\rangle + \mathrm{e}\big(b/2^{i+1}\big)\,|1\rangle}{\sqrt{2}}, \tag{5}$$

that is, ***qubit*** i ***carries the phase*** $2\pi b/2^{i+1}$*.*

Proof. Qubits are processed from the top down, so when qubit i is processed every qubit $l < i$ still holds its input bit b_l. The Hadamard sends $|b_i\rangle$ to $(|0\rangle + \mathrm{e}(b_i/2)\,|1\rangle)/\sqrt{2}$. The gate $CP(\pi/2^{i-l})$ multiplies the $|1\rangle$ component by $\mathrm{e}\big(b_l 2^l/2^{i+1}\big)$, since $\pi/2^{i-l} = 2\pi\,2^l/2^{i+1}$. Collecting the phases,

$$\mathrm{e}\left(\frac{b_i 2^i + \sum_{l<i} b_l 2^l}{2^{i+1}}\right) = \mathrm{e}\left(\frac{b \bmod 2^{i+1}}{2^{i+1}}\right) = \mathrm{e}\left(\frac{b}{2^{i+1}}\right),$$

where the last step uses that the bits above i contribute integers. Qubit i is never touched again. □

$\phi(b)$ is the usual quantum Fourier transform of $|b\rangle$ with the qubit order reversed. The reversal is harmless because this register is never read in Fourier form; the adders below are written for this order, and $\mathrm{QFT}_m^\dagger$ (the gates of QFT_m in reverse order with negated angles) brings the register back.

3.2 Addition as phase rotation (`add`)

Lemma 3.2 (Draper adder)**.** *For an integer a let $\Phi\mathrm{ADD}(a) = \bigotimes_{i=0}^{m-1} P(2\pi a/2^{i+1})$ act on the Fourier register. Then $\Phi\mathrm{ADD}(a)\,\phi(b) = \phi\big((b+a) \bmod 2^m\big)$.*

Proof. $P(\theta)$ multiplies the $|1\rangle$ component by $e^{i\theta}$, so qubit i goes from $\mathrm{e}\big(b/2^{i+1}\big)$ to $\mathrm{e}\big(b/2^{i+1}\big)\,\mathrm{e}\big(a/2^{i+1}\big) = \mathrm{e}\big((b+a)/2^{i+1}\big)$. Adding a multiple of 2^m to $b+a$ changes each exponent by an integer because $i+1 \le m$, so the result is $\phi((b+a) \bmod 2^m)$ by lemma 3.1. □

This is the **Fourier shift theorem** read backwards: multiplying a spectrum by a linear phase shifts the signal. No carry bits and no ancillas are needed; addition of a classical constant is m single-qubit rotations. Subtraction is $\Phi\mathrm{ADD}(-a) = \Phi\mathrm{ADD}(a)^{-1}$. The notebook reduces each angle exactly with integers, $2\pi(a \bmod 2^{i+1})/2^{i+1}$, and omits zero angles.

Example 3.3 ($m = 3$, $b = 3$, $a = 2$)**.** $\phi(3)$ has phases $\mathrm{e}(3/2)\,,\mathrm{e}(3/4)\,,\mathrm{e}(3/8)$ on qubits $0, 1, 2$. $\Phi\mathrm{ADD}(2)$ applies angles $2\pi \cdot 2/2 = 2\pi$ (omitted), π and $\pi/2$. The phases become $\mathrm{e}(5/2)\,,\mathrm{e}(5/4)\,,\mathrm{e}(5/8)$, which is $\phi(5)$.

3.3 Controlled additions (`controlled-add`)

The singly controlled adder replaces each $P(\theta_i)$ by $CP(\theta_i)$ from the control c to qubit b_i:

$$C\text{-}\Phi\mathrm{ADD}(a)\;|c\rangle\,\phi(b) = |c\rangle\;\phi(b + c\,a). \tag{6}$$

Lemma 3.4 (Doubly controlled phase with two CNOTs)**.** *For qubits c_1, c_2 and a target q,*

$$CCP(\theta) = CP(\tfrac{\theta}{2})_{c_2,q}\;CX_{c_1,c_2}\;CP(-\tfrac{\theta}{2})_{c_2,q}\;CX_{c_1,c_2}\;CP(\tfrac{\theta}{2})_{c_1,q}$$

(rightmost gate first). Applied to all targets b_i simultaneously this gives CC-$\Phi\mathrm{ADD}(a)$ with $3m$ controlled phases and only 2 CNOTs.

Proof. On a basis state $|c_1 c_2 q\rangle$ every gate is a phase or a permutation. The total phase exponent is $\frac{\theta}{2}\,q\,\big(c_1 + c_2 - (c_1 \oplus c_2)\big)$, because during the middle gate the second qubit holds $c_1 \oplus c_2$, and the second CNOT restores c_2. Since $c_1 + c_2 - (c_1 \oplus c_2) = 2c_1c_2$, the phase is $\theta\,c_1c_2q$, which is $CCP(\theta)$. All CP gates onto different b_i commute and the CNOTs act only on c_2, so one pair of CNOTs serves the whole register. □

3.4 Controlled modular addition (`controlled-modular-add`)

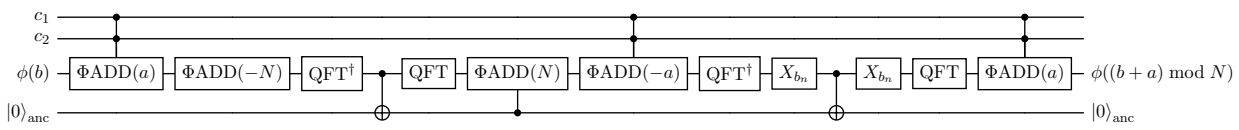


Figure 1: CC-$\Phi\mathrm{ADD}(a)\mathrm{MOD}(N)$ (Beauregard, Fig. 5). The CNOTs read only the top qubit b_n of the $(n+1)$-qubit register.

Theorem 3.5 (Modular adder)**.** *Let $0 \le a, b < N$. The circuit of fig. 1 maps*

$$|c_1c_2\rangle\;\phi(b)\;|0\rangle_{anc} \;\longmapsto\; |c_1c_2\rangle\;\phi\big((b + c_1c_2\,a) \bmod N\big)\;|0\rangle_{anc}\,.$$

Step	Operation	(A) $b+a<N$	(B) $b+a\geq N$	(C) controls off
1	CC-ΦADD(a)	$b+a$	$b+a$	b
2	ΦADD($-N$)	$b+a-N<0$	$b+a-N\geq 0$	$b-N<0$
3	QFT†, $CX(b_n\to\text{anc})$, QFT	anc $=1$	anc $=0$	anc $=1$
4	C-ΦADD(N) on anc	$b+a$	$b+a-N$	b
5	CC-ΦADD($-a$)	$b\geq 0$	$b-N<0$	$b\geq 0$
6	QFT†; sign bit b_n	0	1	0
7	$X(b_n)$, $CX(b_n\to\text{anc})$, $X(b_n)$	anc $=0$	anc $=0$	anc $=0$
8	QFT, CC-ΦADD(a)	$b+a$	$b+a-N$	b

Table 5: State trace of the modular adder. Row 4 already holds the answer; rows 5 to 7 only clean the ancilla.

Proof. By linearity it suffices to follow basis inputs. Three cases cover all of them: (A) $c_1c_2=1$ and $b+a<N$; (B) $c_1c_2=1$ and $b+a\geq N$; (C) $c_1c_2=0$. Table 5 lists the value represented by the b register (as a signed integer) and the ancilla after each step. After an inverse QFT the sign bit b_n equals 1 exactly when the value is negative (section 1).

In every case step 4 holds $(b+c_1c_2a) \bmod N$. In step 6 the sign bit equals $1-\text{anc}$, so step 7 flips the ancilla exactly when it is 1 and returns it to $|0\rangle$. Step 8 undoes step 5. All intermediate values lie in $[-N,2N)$, which the $(n+1)$-qubit register represents without overflow. □

Remark 3.6. Returning the ancilla to $|0\rangle$ is essential, not cosmetic. A leftover ancilla correlated with b would entangle the arithmetic garbage with the counting register and destroy the interference of section 4.

3.5 Modular multiplication

Theorem 3.7 (Controlled multiplier)**.** C-MULT(a)MOD(N) $:=$ QFT$_m^\dagger\cdot\prod_{i=0}^{n-1} CC$-ΦADD($2^ia \bmod N$)MOD($N$)$_{[c,\,x_i]}\cdot$ QFT$_m$ *satisfies, for* $0\leq b<N$*,*

$$|c\rangle\,|x\rangle_n\,|b\rangle_{n+1} \;\longmapsto\; |c\rangle\,|x\rangle_n\,|(b+c\,a\,x) \bmod N\rangle_{n+1}\,.$$

Proof. $ax=\sum_i x_i\,2^ia\equiv\sum_i x_i\,(2^ia \bmod N) \pmod N$. The i-th modular adder is controlled by c and x_i and adds the classical constant $2^ia \bmod N\in[0,N)$. By theorem 3.5 each adder keeps the register in $[0,N)$, so the precondition of the next one holds. □

Intermediate phases in the Fourier domain. After the i-th adder the register is $\phi(B_i)$ with $B_i=\big(b+c\sum_{j\leq i}x_j(2^ja \bmod N)\big) \bmod N$, so qubit l of b carries the phase $\mathrm{e}\big(B_i/2^{l+1}\big)$. Each adder rotates every qubit by $2\pi(2^ia \bmod N)/2^{l+1}$ when $c=x_i=1$, and the reduction mod N is carried out by the sign test of table 5. Only the final QFT$_m^\dagger$ converts the accumulated phases back into the integer $(b+cax) \bmod N$ in the computational basis.

3.6 Controlled U_a: in-place multiplication

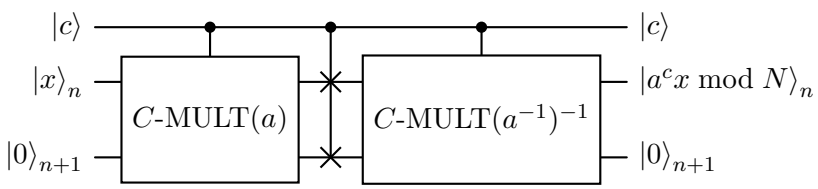


Figure 2: C-U_a (Beauregard, Fig. 7). The controlled swap exchanges x with the low n qubits of b.

Theorem 3.8 (Controlled U_a)**.** *If* $\gcd(a, N) = 1$ *and* $0 \le x < N$*, the circuit of fig. 2 maps* $|c\rangle\,|x\rangle\,|0\rangle\,|0\rangle \mapsto |c\rangle\,|a^c x \bmod N\rangle\,|0\rangle\,|0\rangle$.

Proof. For $c = 0$ all three blocks are the identity. For $c = 1$:

$$|x\rangle\,|0\rangle \xrightarrow{\text{MULT}(a)} |x\rangle\,|ax\rangle \xrightarrow{\text{SWAP}} |ax\rangle\,|x\rangle \xrightarrow{\text{MULT}(a^{-1})^{-1}} |ax\rangle\,|x - a^{-1}(ax)\rangle = |ax\rangle\,|0\rangle\,,$$

all mod N. The last block is the inverse of $|u\rangle\,|v\rangle \mapsto |u\rangle\,|v + a^{-1}u\rangle$, namely $|u\rangle\,|v\rangle \mapsto |u\rangle\,|v - a^{-1}u\rangle$. The inverse $a^{-1} \bmod N$ exists because $\gcd(a, N) = 1$. □

On $\mathbb{Z}_N$ the map $x \mapsto ax$ is a permutation, so U_a is unitary there. The register never leaves $\{0, \dots, N-1\}$ because it starts at 1.

3.7 Modular exponentiation and the entangled state

Write $x = \sum_{j=0}^{t-1} x_j 2^j$. Then

$$a^x = \prod_{j=0}^{t-1} \left(a^{2^j}\right)^{x_j}, \qquad a_j := a^{2^j} \bmod N. \tag{7}$$

The constants a_j are computed classically by repeated squaring, $a_{j+1} = a_j^2 \bmod N$, at a cost of t multiplications. Counting qubit j controls U_{a_j}, which equals $(U_a)^{2^j}$ but is built directly as one multiplier. This is what keeps the circuit polynomial: repeating U_a 2^j times would be exponential. Applying the t controlled multipliers to $|\psi_1\rangle$:

$$|\psi_2\rangle = \frac{1}{\sqrt{M}} \sum_{x=0}^{M-1} |x\rangle\,|a^x \bmod N\rangle_x\,|0\rangle_b\,|0\rangle_{\text{anc}}\,. \tag{8}$$

Residue form. The sequence $a^x \bmod N$ has period r. Write $x = x_0 + kr$ with $0 \le x_0 < r$ and $0 \le k < K_{x_0}$, where $K_{x_0} = \lceil (M - x_0)/r \rceil$. Then

$$|\psi_2\rangle = \frac{1}{\sqrt{M}} \sum_{x_0=0}^{r-1} \Bigg(\sum_{k=0}^{K_{x_0}-1} |x_0 + kr\rangle \Bigg) \otimes |a^{x_0} \bmod N\rangle\,. \tag{9}$$

The state is entangled: it is not a product of a counting state and a work state, because the work value a^{x_0} determines which arithmetic progression ("comb") of x values the counting register holds. Each comb is a periodic pulse train with spacing r.

Eigenbasis form (phase kickback). Define for $s = 0, \dots, r-1$

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \mathrm{e}\!\left(-\frac{sk}{r}\right) |a^k \bmod N\rangle\,. \tag{10}$$

Then $U_a\,|u_s\rangle = \mathrm{e}(s/r)\,|u_s\rangle$, because U_a shifts $k \mapsto k+1$ cyclically and re-indexing the sum produces the factor $\mathrm{e}(s/r)$. Also $\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = |1\rangle$, since $\sum_s \mathrm{e}(-sk/r) = r\,[k \equiv 0]$. Because $U_{a_j} = U_a^{2^j}$ acts on $|u_s\rangle$ as the phase $\mathrm{e}\big(2^j s/r\big)$, each controlled multiplier only writes a phase onto its control qubit, and

$$|\psi_2\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \Big(\frac{1}{\sqrt{M}} \sum_{x=0}^{M-1} \mathrm{e}\Big(\frac{sx}{r}\Big) |x\rangle \Big) \otimes |u_s\rangle\,|0\rangle\,|0\rangle\,. \tag{11}$$

Equations (9) and (11) are the same vector written in two bases of the work register.

Example 3.9 ($N = 21$, $a = 2$)**.** $n = 5$, $t = 10$, $M = 1024$. Repeated squaring gives $a_j = 2, 4, 16, 4, 16, 4, 16, 4, 16, 4$, so all ten multipliers are nontrivial. The orbit of 1 is $1, 2, 4, 8, 16, 11$, so $r = 6$. Since $1024 = 6 \cdot 170 + 4$, the combs with $x_0 \in \{0, 1, 2, 3\}$ have $K = 171$ teeth and those with $x_0 \in \{4, 5\}$ have $K = 170$.

4 Interference: the inverse QFT

4.1 Definition and circuit

The counting register undergoes

$$\mathrm{QFT}_t^\dagger |x\rangle = \frac{1}{\sqrt{M}} \sum_{y=0}^{M-1} \mathrm{e}\Big(-\frac{xy}{M}\Big) |y\rangle \,, \tag{12}$$

which is the unitary DFT matrix (with the engineering sign convention $e^{-j2\pi kn/N}$). In `full` mode it is built as $\lfloor t/2 \rfloor$ swaps, then for $j = 0, \ldots, t-1$ the gates $CP(-\pi/2^{j-l})$ from every $l < j$ onto j followed by H on j. That is $t(t-1)/2$ controlled phases and t Hadamards instead of the M^2 entries of the matrix, the same factorisation that makes the FFT fast.

4.2 Mechanism of constructive and destructive interference

Apply $\mathrm{QFT}_t^\dagger$ to the eigencomponent s of (11):

$$\mathrm{QFT}_t^\dagger \frac{1}{\sqrt{M}} \sum_x \mathrm{e}\Big(\frac{sx}{r}\Big) |x\rangle = \sum_y \alpha_s(y) |y\rangle \,, \qquad \alpha_s(y) = \frac{1}{M} \sum_{x=0}^{M-1} \mathrm{e}(x\,\delta) \,, \quad \delta = \frac{s}{r} - \frac{y}{M}. \tag{13}$$

Each term $\mathrm{e}(x\delta)$ is a unit vector (a phasor) in the complex plane, and consecutive phasors differ by the fixed angle $2\pi\delta$. The geometric series gives

$$|\alpha_s(y)|^2 = \frac{\sin^2(\pi M\delta)}{M^2 \sin^2(\pi\delta)}. \tag{14}$$

- **Constructive.** If δ is an integer, all M phasors point the same way and $|\alpha_s(y)|^2 = 1$.
- **Destructive.** If $M\delta$ is a nonzero integer (not a multiple of M), the phasors are spread uniformly around the circle and sum to exactly zero.
- **Between.** Otherwise the amplitude is concentrated where $|\delta|$ is small.

Proposition 4.1. *If y is the integer nearest to sM/r, then $|\delta| \le 1/(2M)$ and $|\alpha_s(y)|^2 \ge 4/\pi^2 \approx 0.405$.*

Proof. With $u = \pi M|\delta| \le \pi/2$, concavity gives $\sin u \ge 2u/\pi = 2M|\delta|$. Also $\sin(\pi|\delta|) \le \pi|\delta|$. Hence $|\alpha|^2 \ge (2M|\delta|)^2/(M^2\pi^2\delta^2) = 4/\pi^2$. □

Because the $|u_s\rangle$ are orthonormal, the components do not interfere with each other in the counting-register statistics:

$$P(y) = \frac{1}{r} \sum_{s=0}^{r-1} |\alpha_s(y)|^2. \tag{15}$$

4.3 Residue picture and the exact closed form

Starting instead from (9), the comb with offset x_0 maps to

$$\frac{1}{\sqrt{M}} \sum_{k<K_{x_0}} |x_0 + kr\rangle \;\longmapsto\; \sum_y \underbrace{\mathrm{e}\Big(-\frac{x_0 y}{M}\Big)}_{\text{offset} \to \text{phase}} \frac{1}{M} \sum_{k<K_{x_0}} \mathrm{e}\Big(-\frac{kry}{M}\Big) \; |y\rangle \,. \tag{16}$$

The offset x_0 appears only in a phase of modulus one, which vanishes from $|\cdot|^2$. This is the Fourier shift theorem: the unknown starting point of the pulse train is erased, and only its spacing r survives. Different combs are attached to orthogonal work states $|a^{x_0}\rangle$, so their probabilities add.

With $\theta_y = 2\pi r y/M$ and $F(K,\theta) = \left|\sum_{k<K} e^{-ik\theta}\right|^2 = \sin^2(K\theta/2)/\sin^2(\theta/2)$ (equal to K^2 when $\theta \equiv 0$),

$$\boxed{P(y) = \frac{1}{M^2}\Big[\, n_{\rm hi}\, F(K_{\rm lo}+1, \theta_y) + (r - n_{\rm hi})\, F(K_{\rm lo}, \theta_y)\Big]} \tag{17}$$

where $K_{\rm lo} = \lfloor M/r \rfloor$ and $n_{\rm hi} = M \bmod r$ (the number of combs with $K_{\rm lo}+1$ teeth). Summing over y and using Parseval, $\sum_y F(K,\theta_y) = MK$, gives $\sum_y P(y) = (n_{\rm hi}(K_{\rm lo}+1) + (r-n_{\rm hi})K_{\rm lo})/M = 1$. The notebook function `exact_distribution` evaluates (17); it is the zero-noise baseline of the benchmarks. It needs r, which is acceptable for a classical reference and is never used by the circuit.

Example 4.2 ($r \mid M$: $N = 15$, $a = 7$, $t = 8$)**.** Here $r = 4$ and $K = 64$ for every comb. $F(64,\theta_y) = \sin^2(\pi y)/\sin^2(\pi y/64)$ vanishes for every integer y that is not a multiple of 64, and equals 64^2 otherwise. So $P(y) = \frac{1}{4}$ for $y \in \{0, 64, 128, 192\}$ and 0 elsewhere: perfectly sharp peaks.

Example 4.3 ($r \nmid M$: $N = 21$, $a = 2$, $t = 10$)**.** Now $sM/r = 170.67\, s$ is not an integer for $s \notin \{0, 3\}$, so the peaks spread over neighbours. From (17):

y	0, 512	171, 341, 683, 853	170, 342, 682, 854	340, 852, …
$P(y)$	0.1667 each	0.1140 each	0.0285 each	0.0071 each

The six integers nearest sM/r carry 78.9% of the probability, consistent with proposition 4.1.

4.4 The semiclassical IQFT: why $2n+3$ qubits suffice

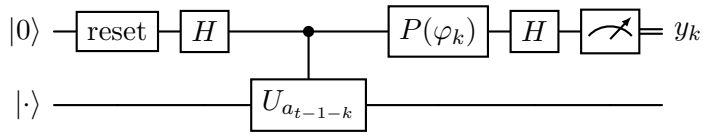


Figure 3: Step $k = 0, \ldots, t-1$ of the semiclassical circuit. The correction $\varphi_k = -\sum_{l<k} y_l\, \pi/2^{\,k-l}$ is applied through classically controlled phase gates (`if_test`) that depend on the bits measured earlier.

Theorem 4.4 (Griffiths and Niu)**.** *The circuit that repeats the step of fig. 3 for $k = 0, 1, \ldots, t-1$, writing y_k into classical bit k, produces the same distribution of $y = \sum_k y_k 2^k$ as the `full` circuit. It uses one control qubit instead of t, hence $2n+3$ qubits in total [4, 2].*

Proof. Start from the `full` circuit. (i) After its initial swaps, position j of the counting register holds the qubit that controlled $U_{a_{t-1-j}}$, and the IQFT processes positions $j = 0, 1, \ldots$ in order: controlled phases from already processed positions, then H. After its H, position j is used only as a control. (ii) A qubit used only as a control may be measured first, with its quantum controls replaced by classical controls on the outcome; the output distribution is unchanged (principle of deferred measurement, and CP is diagonal). (iii) Controlled multipliers on different counting qubits commute, since all U_{a_j} are powers of U_a. So the multiplier controlled by position j may be applied immediately before that position is processed. (iv) After measurement the qubit is in a known basis state; resetting it to $|0\rangle$ and reusing it is equivalent to using a fresh qubit. Steps (i) to (iv) turn the `full` circuit into the circuit of fig. 3 without changing the statistics. □

State at step k. In the eigenbasis, before the second Hadamard the control qubit of component $|u_s\rangle$ is

$$\frac{|0\rangle + \mathrm{e}\big(2^{t-1-k}\tfrac{s}{r} - \sum_{l<k} \frac{y_l}{2^{k+1-l}}\big)\,|1\rangle}{\sqrt{2}} \otimes |u_s\rangle\,. \tag{18}$$

When $s/r = y/M$ exactly, $2^{t-1-k}s/r = y/2^{k+1} \equiv \sum_{l\le k} y_l/2^{k+1-l} \pmod 1$, so the correction leaves the phase $\mathrm{e}(y_k/2) = (-1)^{y_k}$ and H returns $|y_k\rangle$ with certainty. In general, the phases accumulate exactly as in the full IQFT and give the distribution (17).

Example 4.5 ($N = 15$, $a = 7$, $t = 8$)**.** $a_j = 7, 4, 1, 1, \ldots$, so steps $k = 0, \ldots, 5$ use $U_1 = I$: the control qubit stays in $|+\rangle$ and $y_0 = \cdots = y_5 = 0$ with certainty. Step 6 uses U_4 with phase $\mathrm{e}(2s/4) = (-1)^{s_0}$, giving $y_6 = s_0$. Step 7 uses U_7 with phase $\mathrm{e}(s/4) = \mathrm{e}(s_0/4 + s_1/2)$; the correction $-\pi/2$ (applied if $y_6 = 1$) removes $\mathrm{e}(s_0/4)$, leaving $y_7 = s_1$. Hence $y = 64s \in \{0, 64, 128, 192\}$, as in the previous example.

5 Outputs, measurement and classical post-processing

5.1 Collapse

Measuring the counting register (or the t mid-circuit measurements) yields y with probability $P(y)$ from (17). The work register collapses to the normalised state $\propto \sum_s \alpha_s(y)\,|u_s\rangle$, which is not used further. In semiclassical mode the bits arrive least significant first.

5.2 Phase extraction

By proposition 4.1, with probability at least $4/\pi^2$ the outcome y is the integer nearest to sM/r for a uniformly random $s \in \{0, \ldots, r-1\}$, so

$$\left|\frac{y}{M} - \frac{s}{r}\right| \le \frac{1}{2M}. \tag{19}$$

The task is now purely classical: recover the denominator r from the rational number y/M.

5.3 Continued fractions

Algorithm 1: Continued-fraction expansion (`continued_fraction`)

Input: integers $p = y$, $q = M$
Output: coefficients $[c_0; c_1, c_2, \ldots]$ with $y/M = c_0 + 1/(c_1 + 1/(c_2 + \cdots))$

while $q \neq 0$ **do**
  $c \leftarrow \lfloor p/q \rfloor$; append c;
  $(p, q) \leftarrow (q,\ p - c\,q)$; `// Euclid's algorithm`

Algorithm 2: Convergents and period candidate (`convergents`, `find_period`)

Input: coefficients $c_0, c_1, \ldots$; a, N
$h_{-2} \leftarrow 0,\ h_{-1} \leftarrow 1,\ k_{-2} \leftarrow 1,\ k_{-1} \leftarrow 0$;
for $j = 0, 1, \ldots$ **do**
  $h_j \leftarrow c_j h_{j-1} + h_{j-2}$; $k_j \leftarrow c_j k_{j-1} + k_{j-2}$; `// convergent` h_j/k_j
  if $k_j \ge N$ **then stop**: no period found;
  if $a^{k_j} \equiv 1 \pmod N$ **then return** $r \leftarrow k_j$;

Every convergent is in lowest terms, since $h_j k_{j-1} - h_{j-1} k_j = \pm 1$, and the convergents are the best rational approximations of y/M for their denominator size.

Theorem 5.1 (Legendre)**.** *If $|y/M - s/r| < 1/(2r^2)$, then s/r in lowest terms is a convergent of y/M.*

Since $r < N$ and $M \ge N^2$, $|y/M - s/r| \le 1/(2M) \le 1/(2N^2) < 1/(2r^2)$, so the theorem applies. This is the reason for $t = 2n$. The algorithm returns $r' = r/\gcd(s, r)$; this equals r when $\gcd(s, r) = 1$, which happens for a fraction $\varphi(r)/r$ of the values of s. The notebook option `max_multiple` also tests $2r', 3r', \ldots$, which recovers r in most of the remaining cases.

Example 5.2 ($N = 21$, $a = 2$, $y = 853$, $M = 1024$)**.** Euclid: $853/1024 \to c_0 = 0$; $1024 = 1{\cdot}853{+}171$; $853 = 4 \cdot 171 + 169$; $171 = 1 \cdot 169 + 2$; $169 = 84 \cdot 2 + 1$; $2 = 2 \cdot 1$. So $853/1024 = [0; 1, 4, 1, 84, 2]$.

j	c_j	h_j	k_j	test $2^{k_j} \bmod 21$
0	0	0	1	$2 \neq 1$
1	1	1	1	$2 \neq 1$
2	4	4	5	$32 \equiv 11 \neq 1$
3	1	5	6	$64 \equiv 1$: $r = 6$
4	84	424	509	(would stop: $509 \geq 21$)

The large coefficient 84 marks 5/6 as an excellent approximation, just as $[3; 7, 15, 1, 292, \dots]$ marks 355/113 for π.

Example 5.3 (Failures and how they look)**.** For $N = 21$: $y = 341$ gives $[0; 3, 341]$, convergent $1/3$, and $2^3 = 8 \neq 1$, so the plain algorithm fails because $s = 2$ shares the factor 2 with $r = 6$; testing the multiple 6 succeeds. $y = 0$ carries no information. For $N = 15$, $a = 7$: $y = 128$ gives $1/2$ and $7^2 \equiv 4 \neq 1$ ($s = 2$, $r = 4$).

5.4 From the period to the factors

Theorem 5.4. *If r is even and $a^{r/2} \not\equiv -1 \pmod N$, then $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$ are nontrivial factors of N.*

Proof. $N \mid a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1)$. $N \nmid a^{r/2} - 1$, otherwise the order would be at most $r/2$; and $N \nmid a^{r/2} + 1$ by assumption. So the prime factors of N are split between the two factors, and each gcd is a proper divisor. □

For odd N with at least two distinct prime factors and a chosen uniformly among the units, the conditions hold with probability at least $\frac{1}{2}$ [6]. For $N = 15$ they hold for 6 of the 8 units (all except 1 and 14). Example: $N = 21$, $r = 6$, $2^3 = 8 \not\equiv -1$, $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$.

5.5 Success probability per run

Evaluating the post-processing on the exact distribution (17) gives the ideal probability that one run factors N:

Case	plain convergents	with multiples ($m \leq 6$)	uniformly random y (noise floor)
$N = 15,\ a = 7,\ t = 8$	0.500	0.750	0.211
$N = 21,\ a = 2,\ t = 10$	0.331	0.833	0.104

The last column matters for benchmarking: because continued fractions forgive near misses, even a completely random output factors small N a nonzero fraction of the time. Hardware results must be judged against that floor, and the Hellinger fidelity to (17) (floor = 1/64 for $N = 15$, 0.053 for $N = 21$) is the more discriminating metric.

5.6 End-to-end procedure

1. Classical pre-checks (`classical_prechecks`): reject primes; return a factor if N is even, a perfect power, or if $\gcd(a, N) > 1$.
2. Compute n, $t = 2n$, and $a_j = a^{2^j} \bmod N$ by repeated squaring.
3. Build the circuit (`build_shor_circuit`) and run it; each shot yields y.
4. For each y: continued fractions, convergents, first $k_j < N$ with $a^{k_j} \equiv 1$.
5. If r is even and $a^{r/2} \not\equiv -1$, output $\gcd(a^{r/2} \pm 1, N)$; otherwise use another shot or another a.

A Verification of the construction

The notebook's self-test cells check, by exact state-vector simulation in Qiskit: $\Phi\mathrm{ADD}(a)$ for every input on the $(n+1)$-qubit register; $CC\text{-}\Phi\mathrm{ADD}(a)\mathrm{MOD}(N)$ for every $b < N$, a sample of a, and all control values; and $C\text{-}U_a$ for every $x < N$ with the control on and off. A further cell runs the complete circuit noiselessly and compares its output with (17) by Hellinger fidelity.

During development the same functions were also checked with an independent NumPy state-vector simulator (exact branching over mid-circuit measurements): the modular adder for all $a, b < N$ and all control values for $N \in \{15, 21, 35\}$; $C\text{-}U_a$ for all $x < N$ for $N = 15$ ($a \in \{2, 7, 11, 13\}$), $N = 21$ ($a \in \{2, 5, 13\}$) and $N = 35$ ($a \in \{3, 4\}$); and the full output distribution of both modes against (17), with total variation distance below 10^{-12} in every tested case ($N = 15$ with $t \in \{4, 8\}$, $N = 21$ with $t \in \{3, 4, 5\}$, several a, with and without skipping identity multipliers).

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